

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist

#	Code Quality Parameter	Followed	Remarks
1	Consistent Indentation	Yes	Verified via linter runs.
2	Proper File Structure	Yes	Segregated app, models, and training.
3	Meaningful Variable Names	Yes	Variables reflect indicators (e.g., life_exp).
4	Function / Method Names	Yes	Descriptive action names (e.g., init_db).
5	Code Comments	Yes	Docstrings explain methods.
6	Modular Design	Yes	Separated backend REST and UI layers.
7	No Redundant Code	Yes	Shared template structures and style sheets.
8	Error Handling	Yes	Try-except blocks catch prediction errors.

Reusable Components / Modules

#	Component / Module Name	Language / Technology	Where Reused	Reusability
1	SQLAlchemy Database Session Manager	Python (app.py, models.py)	All DB read/write routes	High
2	AJAX POST Request Wrapper	JavaScript (index.js)	All workspace API calls	High
3	SVG Visual Report Builder	Python (app.py)	Report generation endpoint	High

4	Missing Data Mean Imputation	Python (train.py, app.py)	Training and inference pipelines	High
5	Glassmorphism Card Layout Styles	CSS (index.css)	All dashboard panels	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	PEP 8 compliant codebase; clean separation of roles.
Readability	5	Descriptive variable names and complete comments.
Reusability	5	Abstracted styles and modular AJAX elements.
Documentation / Comments	5	Thorough documentation and docstrings.
Overall Score	5	Professional, standard-compliant project layout.